numerically the atomic weight of that element. Thus 1 gramme-atom of hydrogen (H) is 1 gramme, one gramme-atom of oxygen (O) is 16 grammes. It is evident that 1 gramme-atom of any element contains always the same number of atoms—namely, as many atoms as there are molecules in 1 mol. A knowledge of these concepts is very important, for they enable us to count molecules, or atoms, by weighing them, as it were, and for this reason it is absolutely necessary to know the exact number of molecules contained in a mol. As we have stated before, this number was first computed correctly—as regards the order of magnitude at least—by Loschmidt in 1865. But the first really reliable calculation was made in 1900 only, on the basis of Planck's radiation law. To-day, the most reliable numerical value of this important Loschmidt's number (L) is: $6{\cdot}024 \times 10^{23}$.

This means that 1 mol of a substance—for instance, 32 grammes of oxygen gas—contains almost 1 quadrillion molecules.

Furthermore, Loschmidt's number gives us an exact knowledge of the masses of the individual atoms and molecules in terms of the customary unit of 1 gramme. Since 1 mol of hydrogen gas weighs 2 grammes, we can easily find—dividing by Loschmidt's number—that the mass of a hydrogen molecule (H_2) is $3{\cdot}34 \times 10^{-24}$ gramme, and therefore, the mass of a hydrogen atom (H) is $1{\cdot}67 \times 10^{-24}$ gramme. The masses of all other atoms and of all kinds of molecules (when their atomic structure is known) can be computed.

Now let us turn our attention to the magnitude of the electric charge which is linked, through electrolysis, with an atom or molecule—the atom of electricity, the electron. We have already stated that a certain quantity of electricity, namely:

$$F = 96{,}520 \text{ coulombs}$$

is always linked to 1 gramme-atom of a univalent substance. But it is very difficult to appreciate the magnitude of this charge. It is much greater than any electric charge that can be produced in any single substance in the laboratory. If both the moon and the earth were to carry, each, a charge of this magnitude, they would attract or repel each other—despite the great distance between them—with a force equivalent to several hundreds of kilogrammes. This is the charge carried by 1 gramme-atom of a univalent substance. But since 1 gramme-atom always contains the same number of atoms (Loschmidt's number), the charge carried by an individual univalent atom is determined by dividing the equivalent charge F by Loschmidt's number; this